

Windows Endpoint Analysis

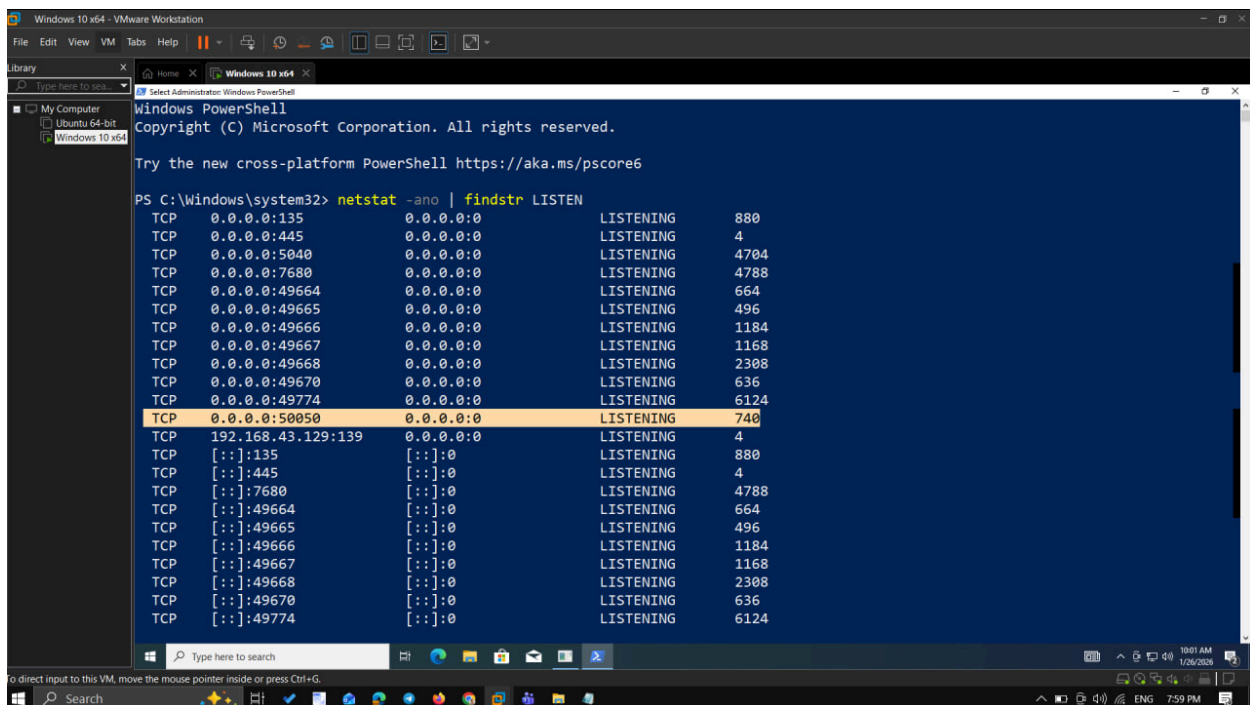
You've been assigned to remotely investigate a compromised Windows workstation as part of a live incident response scenario. The affected system has been imaged and all necessary forensic artifacts have been gathered. Your objective now is to investigate the scope of the compromise and identify any persistence mechanisms installed by the attacker.

<https://challenges.malwarecube.com/#/c/592319ba-7fd3-4322-ad14-1fa946fedc46>

1) Identify suspicious listening network activity.

Malware and backdoors often open listening ports to receive commands or establish persistence. Enumerating listening ports helps identify unauthorized network exposure.

```
netstat -ano | findstr LISTEN
```



- Ports like 135, 445, 139 were excluded as legitimate Windows services.
- Dynamic RPC ports were excluded unless linked to suspicious processes.

Suspicious listening port: 50050

2) Identify the process associated with the suspicious port and Analyze loaded modules of the suspicious process.

Once a suspicious port is identified, the owning process must be investigated to determine if it is malicious.

Reviewing loaded DLLs helps confirm malicious intent, especially network-related capabilities.

```
tasklist | findstr 740  
  
tasklist /m /fi "PID eq 740"
```

- Used the PID (740) identified from netstat.
- Queried running processes matching the PID.
- Core Windows DLLs were excluded as normal.
- Network DLLs strengthened the suspicion but were not standalone proof.

```
Select Administrator: Windows PowerShell

PS C:\Windows\system32> tasklist | findstr 740
svchost.exe                2740 Services                0      13,724 K
challenge.exe              740 Console                  1       4,060 K

PS C:\Windows\system32> tasklist /svc | findstr 740
svchost.exe                2740 CryptSvc
challenge.exe              740 N/A

PS C:\Windows\system32> tasklist /m /fi "PID eq 740"

Image Name                  PID Modules
=====
challenge.exe              740 ntdll.dll, KERNEL32.DLL, KERNELBASE.dll,
ADVAPI32.dll, msvcrt.dll, sechost.dll,
RPCRT4.dll, WS2_32.dll, mswsock.dll

PS C:\Windows\system32>
```

Process name: `challenge.exe`

Identified network-related modules such as:

`msvcrt.dll`
`mswsock.dll`

3) Identify the malware's parent process name.

Parent process analysis reveals how the malware was executed (user-driven vs system-driven).

- Identified ParentProcessId as **2192**.
- Queried PID 2192.

```
Select Administrator: Windows PowerShell
PS C:\Windows\system32> Get-CimInstance Win32_Process | Where-Object {$_.ProcessId -eq 740} | Select Name, ParentProcessId

Name          ParentProcessId
-----
challenge.exe      2192

PS C:\Windows\system32> Get-Process -Id 2192

Handles  NPM(K)  PM(K)  WS(K)  CPU(s)  Id  SI ProcessName
-----
2089      77    51748  117100  9.53    2192  1 explorer

PS C:\Windows\system32>
```

Parent process ID : 2192

Parent process: explorer.exe

4) Check for malicious network shares.

Attackers may create hidden or custom network shares for lateral movement or data exfiltration.

```
net share
```

```
Select Administrator: Windows PowerShell

PS C:\Windows\system32> net share

Share name      Resource                                Remark
-----
C$              C:\                                    Default share
IPC$            C:\Windows                            Remote IPC
ADMIN$          C:\Windows                            Remote Admin
xkalibur        C:\Users\maro\AppData\Local\Temp\46d5b8556d0d3e30ec1

The command completed successfully.

PS C:\Windows\system32>
```

C\$, ADMIN\$, IPC\$ excluded as default Windows shares.

Malicious share name: xkalibur

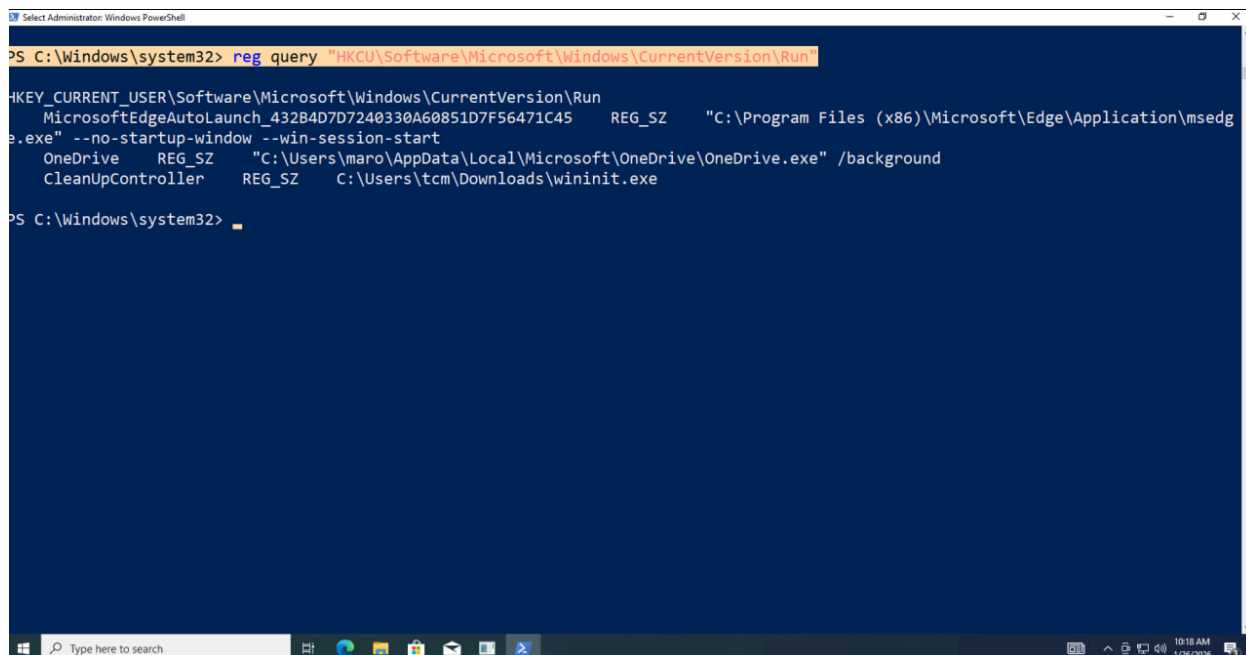
Path: C:\Users\maro\AppData\Local\Temp\46d5b8556d0d3e30ec1

5) Identify persistence via Registry Run keys.

Registry Run keys allow malware to execute automatically upon user login.

```
reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run"
```

- Microsoft Edge and OneDrive excluded as legitimate.
- Executable running from Downloads directory flagged.



```
PS C:\Windows\system32> reg query "HKCU\Software\Microsoft\Windows\CurrentVersion\Run"

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
    MicrosoftEdgeAutoLaunch_432B4D7D7240330A60851D7F56471C45    REG_SZ    "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --no-startup-window --win-session-start
    OneDrive    REG_SZ    "C:\Users\maro\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background
    CleanUpController    REG_SZ    C:\Users\tcm\Downloads\wininit.exe

PS C:\Windows\system32>
```

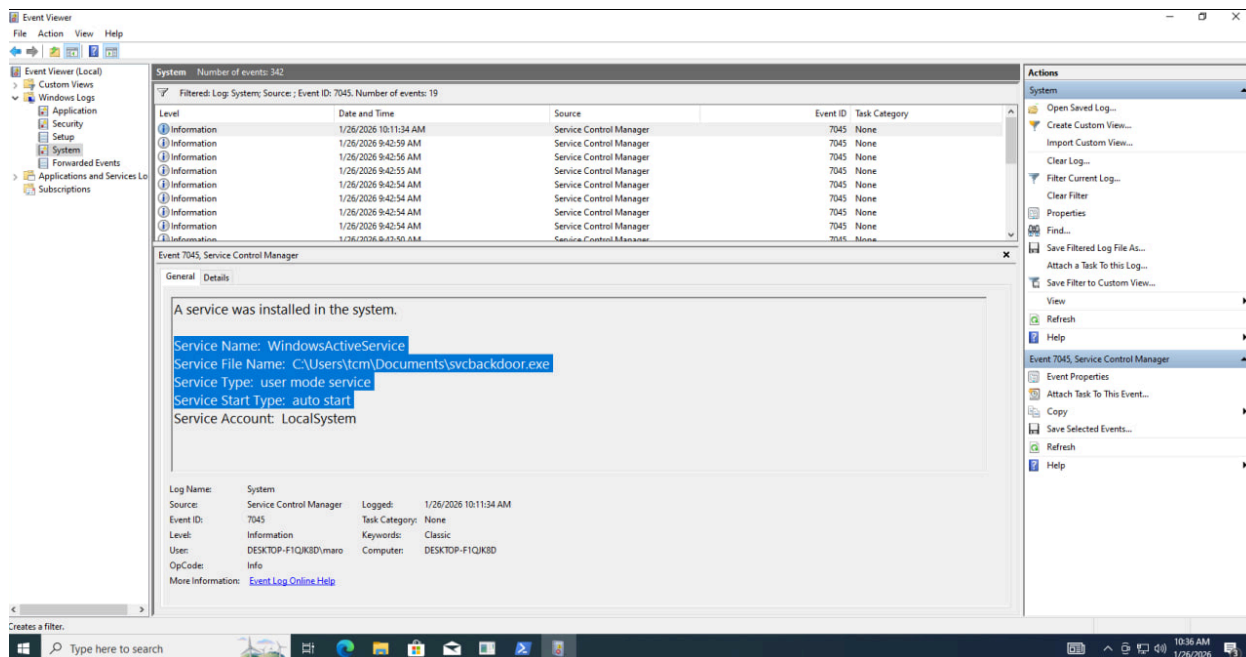
Suspicious Run Key: `CleanUpController`

Executable Path: `C:\Users\tcm\Downloads\wininit.exe`

6) Identify malicious service installed by attacker.

Services provide high-privilege, auto-start persistence. Service creation is logged as Event ID 7045.

- Event Viewer → Windows Logs → System
- Filter by **Event ID: 7045**



Service Name: WindowsActiveService

START_TYPE: Auto Start

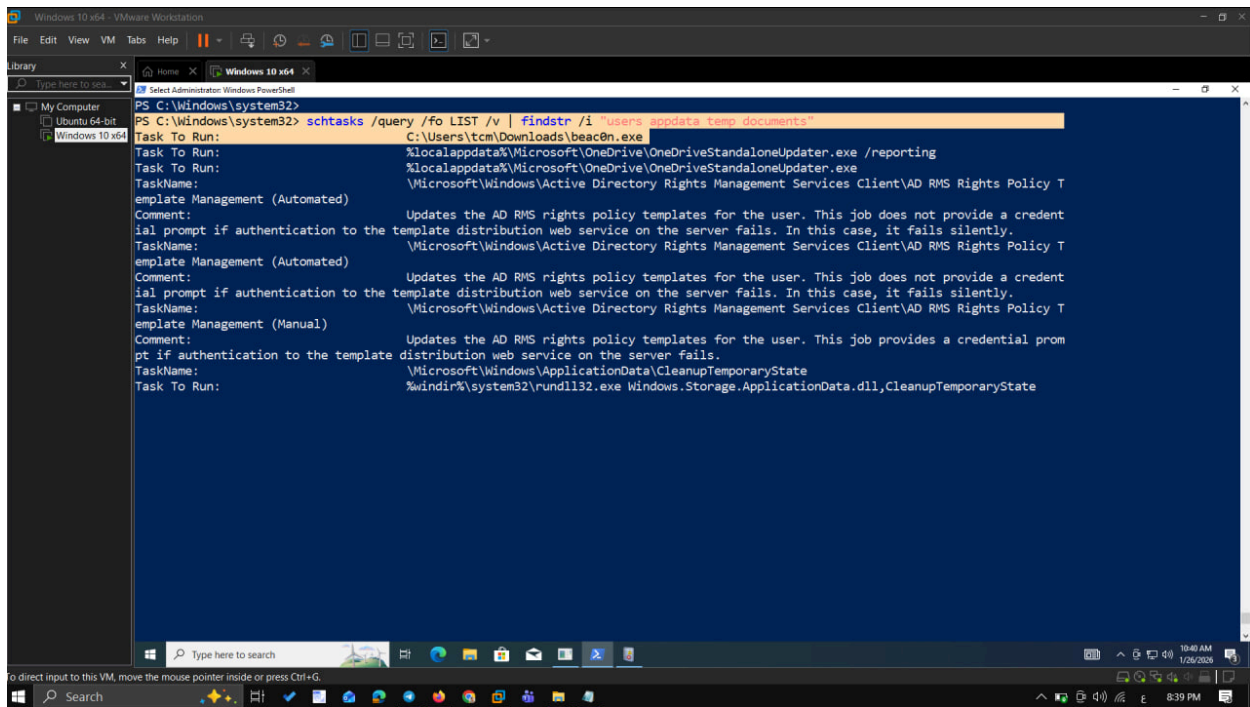
Full path of the service: C:\Users\tcm\Documents\svcbackdoor.exe

7) Identify malicious scheduled task.

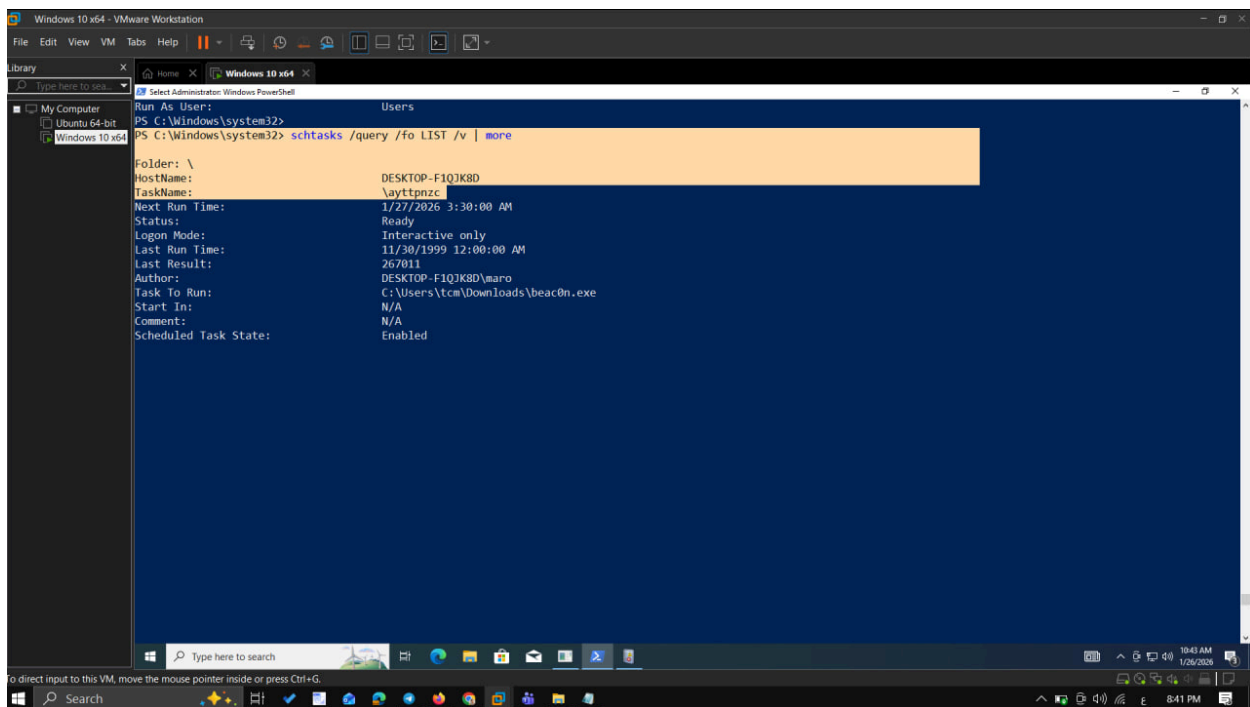
Scheduled tasks are commonly used as redundant persistence mechanisms.

- Enumerated all scheduled tasks.
- Looked for:
 - Random task names
 - Tasks outside Microsoft folders

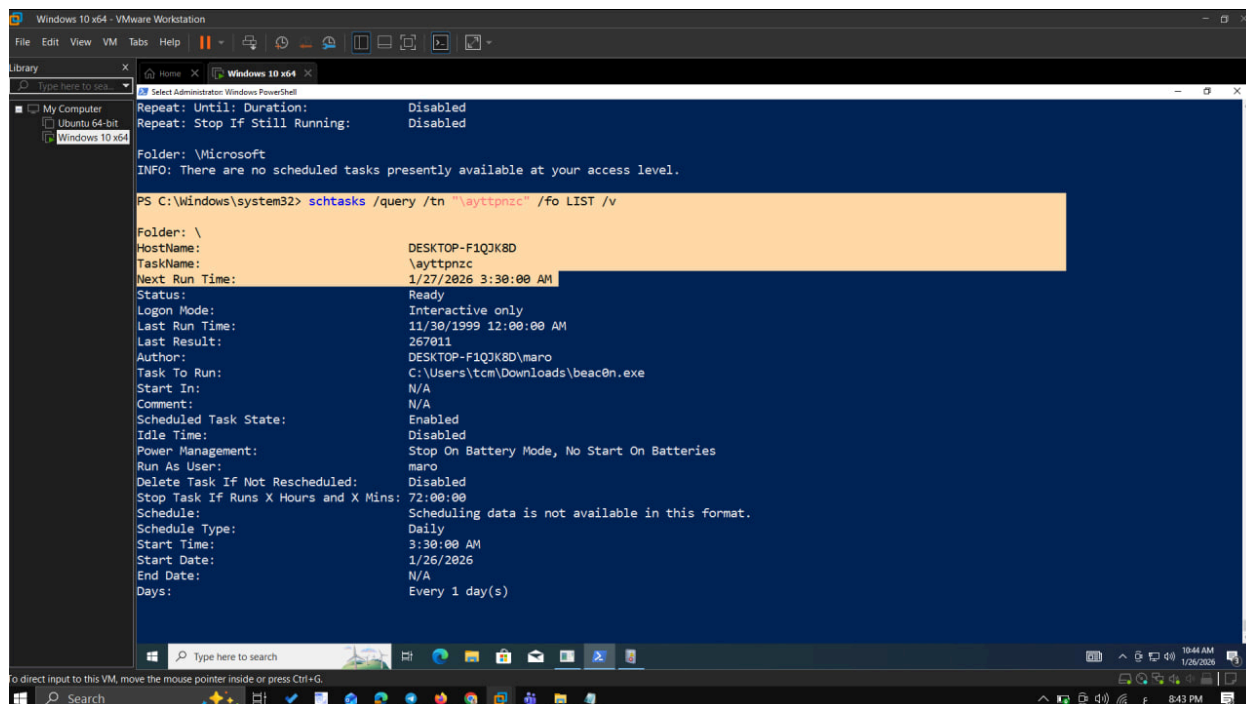
```
schtasks /query /fo LIST /vschtasks /query /fo LIST /v | find
str /i "users appdata temp documents"
```



`schtasks /query /fo LIST /v | more`




```
schtasks /query /tn "\ayttpnzc" /fo LIST /v
```



Payload / Malware binary: `C:\Users\tcm\Downloads\beac0n.exe`

Malicious Task Name: `\ayttpnzc`

Time to run : `03:30:00`

MITRE ATT&CK Matrix (Enterprise – Windows)

Initial Access

- **T1566 – Phishing**

The attacker gained initial access through a suspicious email that led to user interaction and execution of a malicious file.

Execution

- **T1204.002 – User Execution: Malicious File**

The malware (`challenge.exe`) was manually executed by the user on the compromised endpoint.

Persistence

- **T1547.001 – Boot or Logon Autostart Execution: Registry Run Keys**

A malicious Run key entry (`CleanUpController`) was created under:

`HKCU\Software\Microsoft\Windows\CurrentVersion\Run`

- **T1543.003 – Create or Modify System Process: Windows Service**

A malicious Windows service (`WindowsActiveService`) was installed to ensure persistence and execution at system startup.

- **T1053.005 – Scheduled Task/Job: Scheduled Task**

A scheduled task (`ayttpnzc`) was created to periodically execute a malicious binary (`beac0n.exe`).

Defense Evasion

- **T1036 – Masquerading**

Malicious files were named to resemble legitimate Windows binaries (e.g., `wininit.exe`) to evade detection.

Discovery

- **T1135 – Network Share Discovery**

An unauthorized network share (`xkalibur`) was created, enabling access to local system resources.

Command and Control

- **T1071.001 – Application Layer Protocol: Web Protocols**

The malware loaded `WS2_32.dll` and `mswsock.dll` , confirming the use of Windows networking APIs for command-and-control communication.

- **T1571 – Non-Standard Port**

The malware opened a listener on TCP port **50050**, which is a non-standard port commonly used to evade basic network monitoring.

Exfiltration (Potential)

- **T1041 – Exfiltration Over C2 Channel**

The presence of an active backdoor listener indicates the potential capability to exfiltrate data over the established C2 channel.

Conclusion

This investigation confirmed that the Windows host was compromised and that the attacker established multiple persistence mechanisms to maintain access. A malicious executable was found listening on a non-standard port, indicating backdoor functionality.

Persistence was achieved using an auto-start Windows service, a malicious scheduled task with a randomized name, and a registry Run key. The attacker also created a custom network share, likely for staging or data transfer.

All malicious artifacts were executed from user-writable directories rather than trusted system locations, which strongly indicates malicious activity. Overall, the findings show a fully compromised system with layered persistence techniques in place.